

Chapter 1

INTRODUCTION TO EXPERIMENTATION

1.1 Why Conduct Experiment

In our everyday lives, we can observe numerous phenomena in our surroundings. For example, the blue color of the sky, the swinging of a wall clock's pendulum, the fact that a cyclist does not fall despite the minimal contact area between the bicycle tire and the road surface, and so forth.

Other phenomena can only be observed through the use of specialized instruments in a laboratory setting for instance, interference patterns resulting from a double slit, electron diffraction, and others. The aim of physics is to seek ways to explain or interpret such observed phenomena. The key to discovering these explanatory paths lies in experimentation.

Since the case of the bicycle may be too complex to analyze and understand all at once, we begin with a simpler case: understanding the fundamental properties governing a rotating wheel just one, to start with. This endeavor can be carried out in a laboratory, where we have the freedom to observe the environment closely, to construct models that help us comprehend the actual situation, and so forth.

Modern science is founded upon observation and experimentation. Therefore, a physicist must be trained (and must train oneself) earnestly in order to become a keen observer and a capable experimenter.

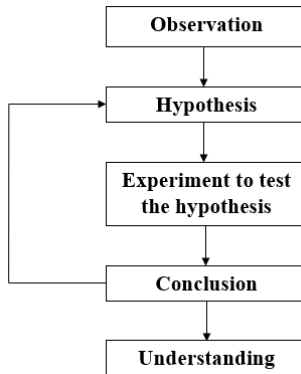


Figure 1.1: The process from Observation to Understanding

1.2 From Observation to Understanding

In general, the progression from observation to drawing conclusions can be illustrated in several steps or phases (see Figure 1.1). We will describe these steps through a simple example.

Let us assume that we know nothing about physics. We observe a container of water with a wooden ball, a rubber ball, and a piece of aluminum shaped like a cube inside it. Our observation tells us that the wooden ball and rubber ball float, while the aluminum cube sinks.

Science begins to develop when humans are motivated to understand these phenomena. To "understand" means being able to formulate laws or principles that explain (or "satisfy") these events and can predict new occurrences.

In the case of the behavior of the balls and the cube in water, for example, one might formulate a principle that states: "Round objects float, and cube-shaped objects sink." This statement is called a hypothesis because it has not yet been

proven. In this case, we must realize that what the hypothesis proposes covers a broader field than what has been observed.

We have not yet observed that all round objects float we have only seen that the wooden and rubber balls do. In this hypothesis, we are generalizing the observed behavior of the wooden and rubber balls into a broader statement about all round objects. To make this hypothesis more credible, further experiments must be conducted. We need to design and carry out an experiment to test the validity of the hypothesis.

Suppose we take a round cork and a cube-shaped stone and drop both into water. Once again, we observe that the round object floats and the cube sinks. At this stage of experimentation, there is no reason to reject the hypothesis; on the contrary, this new experiment strengthens our confidence in it. However, if someone happens to throw an aluminum ball into the water, and we observe that it sinks, then we must discard the original hypothesis and consider a new hypothesis that is capable of explaining all the events observed thus far. Based on this revised hypothesis, we design and carry out new experiments to test its validity, and so the process continues. It is, in a sense, a cycle that we must undergo until we arrive at a hypothesis that is consistent with a wide range of experiments. Such a hypothesis gradually evolves into a scientific law.

This simple example illustrates the fact that what we call science is a continuous chain of events beginning with a hypothesis, followed by experimentation, then leading to a new hypothesis, further experimentation, and so on. Eventually, through this sequence, it may become possible to formulate a scientific law. Such laws, along with the relationships between measurable quantities, can often be expressed through the aid of a model or a theory (such as the wave theory or quantum

theory). With such theories, we are able to make quantitative predictions about new phenomena, and, if necessary, revise the theory accordingly.

1.3 Example from History

To illustrate the process by which scientific laws are established, we will present a brief account of the experiments and ideas from the 19th century concerning the nature of heat. This story is a summary of Chapter 4 of the book “How to Know” (bagaimana cara mengerti) written by Goldstein. The chapter is titled: *Is Heat a Substance?*

At the end of the 18th century, there were two prevailing theories about heat. The first theory, which was supported by most scientists at the time, proposed that heat is a substance that can move from one object to another. Several known phenomena could be explained using this hypothesis. For example, the expansion of objects when heated was accounted for as follows: the heat substance, upon entering the object, surrounds its atoms, thereby increasing the space between them and causing the object to expand. This hypothetical heat substance was called **caloric**, and the theory describing its properties was known as the **caloric theory**.

The second theory proposed that heat is not a substance, but rather the motion of the particles within the heated object. This theory was based on the observation that motion generates heat. Although it has now been conclusively proven that the caloric theory is incorrect, several important properties of heat were nonetheless discovered through its framework. Even today, we still speak of a “flow of heat,” despite the fact that “flowing” is a behavior typically associated with substances. We now understand that heat is the motion of atoms; however,

many thermal phenomena are still more easily conceptualized through the “caloric approach,” where heat is treated as though it were a fluid. The thermometer though it may appear simple to us now proved to be a powerful tool in helping to explain thermal phenomena. Before the invention of the thermometer, the “degree of heat” of an object was “gauged” by simply touching it. An English scientist, Francis Bacon, compiled a list of natural situations in which, according to him, heat was present. This list included the following:

- Rays come from the sun
- Liquids that are boiling or being heated
- All forms of open flame
- Any object that is vigorously rubbed
- Soft wool, fine feathers, and furry animal skins
- Aromatic and “hot” herbs (which create a stinging sensation in the mouth, such as chili peppers)
- Strong vinegar and all concentrated acids (since these liquids “burn” the eyes and sensitive skin)

We now understand, for instance, that wool feels warm because it acts as an insulator, not because it contains heat. In his attempt to understand heat, Francis Bacon performed something very important in the scientific process: he classified observed and experienced phenomena. The fact that his method of classification turned out to be incorrect was due to the inadequacy of his “measuring instrument” (namely, the human sense of touch. Had he possessed a thermometer (which, unfortunately, had not yet been invented at the time), he might have developed a different classification.

It was only after the invention of the thermometer around the year 1600, that heat could be measured quantitatively and

objectively. The principle of **thermal equilibrium** was established, which states: when two objects at different temperatures are brought into contact, they will, after some time, reach the same temperature. This final temperature lies somewhere between the initial temperatures of the two objects. This principle significantly strengthened confidence in successive hypothesis. To clearly distinguish between the concepts of the amount of heat and the degree of temperature, a commonly used and easily understood analogy was introduced. Imagine two cylinders of different diameters, each filled with water in different volumes. The two cylinders are connected by a pipe with a valve. Initially, the valve is closed. The water levels in the two cylinders are not the same (see Figure 1.2(a)).

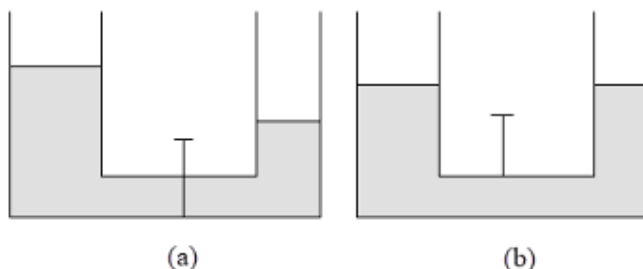


Fig 1.2: (a) Valve closed, (b) Valve open

Once the valve is opened, the water "seeks a new level." The water level in the wider container changes only slightly, while in the narrower cylinder, the change in height is much more noticeable. The height of the water can be compared to temperature, whereas the volume of water in each vessel represents differing quantities. Explaining this phenomenon using atomic motion is considerably more difficult. For those

who supported the theory that "heat is a substance," they also had to accept the idea that the amount of heat is conserved. According to this line of thinking, heat cannot be created or destroyed (see the simple models in Figures 1.2(a) and 1.2(b)). No water is added or removed; the total amount of water in the two cylinders shown in Figures 1.2(a) and 1.2(b) remains the same.

Further experiments on heat were carried out. It was discovered that if two equal amounts of water at different temperatures (for example, 0°C and 100°C) are mixed, the final temperature of the mixture lies midway between the two, that is, 50°C . Then, a similar experiment was performed, this time using equal masses of mercury (at 100°C) and water (at 0°C). The final temperature turned out not to be 50°C , but only 3°C . When the same experiment was conducted with equal volumes instead of equal masses, the final temperature was still not 50°C , but again remained close to the initial temperature of the water.

This phenomenon led physicists to the concept of heat capacity. It became clear that the heat capacity of water is higher than that of mercury. As a result, a more quantitative understanding of the amount of heat was developed. If the temperature of one kilogram of water is raised by 10°C , it is said to have absorbed a certain amount of heat. If the temperature increase is 20°C , the heat absorbed is twice as much. Therefore, the amount of heat Q can be expressed as: $Q = C \times (t_2 - t_1)$, with $(t_2 - t_1)$ is the temperature change and C is a constant related to the heat capacity. The value of C can be chosen arbitrarily. A convenient choice is to assign the value of 1 (one) to the heat capacity of water. Then, the principle of conservation of heat can be applied to calculate the heat capacity of mercury. The idea is straightforward: the

amount of heat lost by the mercury equals the amount of heat gained by the water. Therefore,

$$C_{\text{mercury}} \times (100^{\circ}\text{C} - 3^{\circ}\text{C}) = C_{\text{water}} \times (3^{\circ}\text{C} - 0^{\circ}\text{C})$$
$$C_{\text{mercury}} = 0,03 \times C_{\text{water}}$$

Thus, the heat capacities of other substances could be determined using the same method that had been applied to mercury. Eventually, this also enabled scientists to make predictions regarding final temperatures. For example, if 1 kg of mercury at 100°C is brought into contact with 1 kg of copper at 0°C , a prediction about the final temperature can be made. These predictions were then tested through experiments, and the results were found to be consistent with the predictions. All of this further strengthened confidence in the concepts of heat conservation and in the idea of heat as a substance.

Scientists, however, began to raise logical objections to this experimental concept specifically, the notion that a lighter substance such as water has a higher heat capacity than a heavier one like mercury. Their reasoning was as follows: if heat is actually a form of motion (as some scientists proposed), then the heavier the atoms, the more effort or energy would be required to move them. Therefore, heavier substances should logically possess higher heat capacities, not lower ones.

Many more experiments were conducted not only concerning final temperatures (i.e., thermal equilibrium), but also involving the rate at which heat flows. All results at the time, however, could still be interpreted within the framework of the caloric theory the idea that heat is a conserved fluid-like substance.

Then came a turning point with the experiments carried out by Count Rumford during the late 1700s and early 1800s. Looking back at his experiments today, we can clearly conclude

that his findings strongly supported the hypothesis that heat is a form of motion.

Nonetheless, it would take another 50 years before the caloric theory was finally abandoned and the kinetic theory of heat was fully accepted. Prince Rumford conducted numerous experiments. We will discuss just three (3) of them:

1. He attempted to measure the mass of heat.
2. He demonstrated that friction appears capable of generating an unlimited amount of heat.
3. He showed that if a salt solution is overlaid with a layer of pure water, the salt solution will slowly rise into the pure water, without any stirring.

1.3.1 Does Heat Have Weight

If heat is indeed a substance, it must have mass. If from an experiment we are able to conclude that heat has mass, we must accept the view that heat is a substance. But “unfortunately,” if on the contrary we are not able to detect the mass originating from heat, we cannot discard the hypothesis of heat as a substance, because there is always uncertainty in the result of weighing. At most, we can only say that if heat truly has weight, then the value of this weight is smaller than the upper limit of the uncertainty. We can learn a lot from the story of how Rumford tried to eliminate the possible errors that might exist.

We will not discuss it in detail, but only conclude that Rumford (within the limits of uncertainty) did not find any indication of the existence of mass in heat. Rumford ended his report with: “all efforts to weigh heat, until its effects become apparent on the balance, will be in vain.” From Rumford’s statement, we draw a simple conclusion: if heat has mass, then this mass is smaller than the mass that can still be detected with

Rumford's balance. In any case, it began to be generally accepted that heat has no mass. This view was even accepted by supporters of the caloric theory.

1.3.2 Friction Generates Heat

In his duty as a supervisor of cannon manufacturing, Rumford noted that "boring holes" generated heat. The proponents of the caloric theory proposed several explanations. The boring process transforms a large piece of metal into small fragments. Their hypothesis was that perhaps the metal, when cut into small pieces, has a lower heat capacity than when in a large chunk. This would then result in a release of heat during the boring process. Another hypothesis suggested that heat was produced by a chemical reaction between the metal and air during boring. Rumford tested both of these hypotheses.

He measured the heat capacity of the metal in the form of small fragments and in the form of a large chunk, and found that the values were the same. Thus, the first hypothesis could be dismissed. As for the second hypothesis, he conducted the following experiment: the entire system was submerged in water. And Rumford obtained astonishing results, so much heat was produced during the boring process that the water began to boil after a while. The heat that raised the temperature of the cannon could not have come from the water, because the water itself was being heated. (From the principle of heat conservation, the water should actually cool down if the cannon was warming up.) Rumford also noted that there was no decomposition of the water. Rightly, Rumford concluded that the amount of heat is not conserved and that heat is related to motion. This concept was not immediately accepted, because many phenomena could still be explained by the caloric theory, while the new concept had yet to be clearly formulated. It was only 50 years later that the concept of heat as motion was accepted by scientists in general, namely, after Joule

established the relationship between the work done and the heat generated.

1.3.3 Atomic Motion

The third experiment conducted by Rumford was as follows. He prepared a layer of salt solution in a glass container. On top of this layer, he carefully placed a layer of pure water, which has a lower density than the salt solution. This placement was done with great caution to avoid mixing. If all the molecules were truly at rest, then the salt solution would continuously remain at the bottom. However, if the molecules were in motion, the salt molecules would gradually diffuse into the pure water layer, resulting in mixing. Rumford observed that mixing occurred and proposed three hypotheses:

1. There is a kind of attractive force in the pure water that pulls the salt upward.
2. Because of slight drafts in the experimental room, temperature fluctuations occurred, which caused convection in the liquid.
3. There is continuous molecular motion.

Unfortunately, Rumford did not take further action on this experiment. In fact, the experiment had already provided a clear indication of what Rumford should have done as a follow-up, namely, he should have observed the diffusion phenomenon at different temperatures. Rumford would then have noted that the rate of diffusion changes significantly with varying temperatures.

The story about heat is an example of how science develops. The crucial steps are observation, testing the hypothesis, in order to determine the next course of action, which is: to create an entirely new hypothesis or to refine the existing one. And so it goes on continuously. This cycle is what is important for physicists, and in this regard, you will be trained during the practicum sessions.

1.4 Linear Regression Method

In the world of physics, we are often confronted with sets of experimental data obtained through a series of observations or experiments. To understand the relationships between the observed variables, one of the most commonly used statistical methods is linear regression. Linear regression is an approach used to model the relationship between a dependent variable and one or more independent variables, under the assumption that the relationship between the variables is linear in nature.

The basic concept of linear regression is to find a straight line (in the case of simple linear regression) or a flat plane (in multiple linear regression) that best fits the set of existing data points. In the context of physics, this line or plane represents the mathematical model of the relationship between the observed variables. For instance, the relationship between temperature and the change in volume of a particular gas, or the relationship between speed and frictional force acting on an object.

An understanding of linear regression is very important in physics laboratory work because it not only allows us to comprehend the relationships between variables, but also provides insight into the precision and accuracy of our observations. Through residual analysis and other parameters, we can evaluate how well our linear model represents the data and how large the possible errors might be in our measurements. Thus, linear regression becomes a highly valuable tool in the analysis and interpretation of experimental data in physics.

1.4.1 Linear Regression Formula

The linear relationship between the dependent variable y and the independent variable x is given by the equation $Y = A + BX$. This equation is a mathematical model to describe the data pairs of variables x and y , namely (x, y) . Figure 1.3 displays

the points (x, y) and the linear line $Y = A + BX$. For one value of the independent variable x , there are two values on the vertical coordinate, namely y , which is the pair of x , and Y , which is the result of calculation from the equation $Y = A + BX$, namely the value y on the straight line. Here $x = X$. The difference between y and Y is the deviation of the point (x, y) from the linear line $Y = A + BX$, which can be written as:

$$\delta = y - Y \quad (1.1)$$

The smaller the δ , the better the mathematical model $Y = A + BX$ describes the data set (x, y) . This mathematical model, or equation, is what will be sought by calculating or determining the values of A and B that will yield the most probable value of Y . If Y is the most probable value, then the sum of the squared deviations, namely, $\sum_{i=1}^N \delta_i^2 = \sum_{i=1}^N (y_i - A - Bx_i)^2$, is the smallest (minimum).

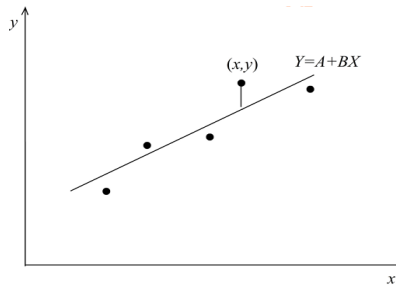


Figure 1.3: A mathematical model, $Y = A + BX$ representing the linear relationship between y and x

Thus, the least squares principle for linear fitting can be expressed with the following condition:

$$\sum (\delta_i)^2 \rightarrow \text{minimum} \quad (1.2)$$

can be written as:

$$\sum [y_i - (A + Bx_i)]^2 = M \quad (1.3)$$

The condition that M is minimum is:

$$\frac{\partial M}{\partial A} = 0 \quad (1.4)$$

and

$$\frac{\partial M}{\partial B} = 0 \quad (1.5)$$

First, we find

$$\begin{aligned} (\delta_i)^2 &= [y_i - (A + Bx_i)]^2 \\ &= y_i^2 - 2y_i(A + Bx_i) + A^2 + B^2x_i^2 \\ &= y_i^2 - 2y_iA - 2y_iBx_i + A^2 \\ &\quad + B^2x_i^2 \end{aligned} \quad (1.6)$$

and,

$$\begin{aligned} M &= \sum (\delta_i)^2 \\ &= \sum y_i^2 - 2A \sum y_i - 2B \sum x_i y_i + \sum A^2 \\ &\quad + B^2 \sum x_i^2 \end{aligned} \quad (1.7)$$

Next, we differentiate M with respect to A and B and apply the minimum condition,

$$\frac{\partial M}{\partial A} = 2NA + 2B \sum x_i - 2 \sum y_i = 0 \quad (1.8)$$

$$\frac{\partial M}{\partial B} = 2B \sum x_i^2 + 2A \sum x_i - 2 \sum (x_i y_i) = 0 \quad (1.9)$$

Thus, we obtain two simultaneous equations with two unknown parameters, namely A and B , which we will solve using the matrix method. These two simultaneous equations are written in matrix form as follows,

$$\begin{pmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} \sum y_i \\ \sum (x_i y_i) \end{pmatrix} \quad (1.10)$$

with:

$$\Delta = \begin{vmatrix} N & \sum x_i \\ \sum x_i & \sum x_i^2 \end{vmatrix} = N \sum x_i^2 - \left(\sum x_i \right)^2 \quad (1.11)$$

Finally, A and B are obtained as follows,

$$A = \frac{\begin{vmatrix} \sum y_i & \sum x_i \\ \sum (x_i y_i) & \sum x_i^2 \end{vmatrix}}{\Delta} = \frac{\sum x_i^2 \sum y_i - \sum x_i \sum (x_i y_i)}{\Delta} \quad (1.12)$$

$$B = \frac{\begin{vmatrix} N & \sum y_i \\ \sum x_i & \sum (x_i y_i) \end{vmatrix}}{\Delta} = \frac{N \sum (x_i y_i) - \sum x_i \sum y_i}{\Delta} \quad (1.13)$$

Equation (1.12) and equation (1.13) provide the best estimates for A and B in the straight-line equation $Y = A + BX$ from a set of N data pairs $(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)$. This line is called the least squares fit to the data, or also referred to as the linear regression line of y on x .

The errors for A and B are obtained by applying error propagation to Equations (1.12) and (1.13), resulting in

$$\sigma_A = \sigma_y \sqrt{\frac{\sum x^2}{\Delta}} \quad (1.14)$$

and

$$\sigma_B = \sigma_y \sqrt{\frac{N}{\Delta}} \quad (1.15)$$

with Δ as given in equation (1.11), and σ_y given by the equation

$$\sigma_y = \sqrt{\frac{1}{N-2} \sum_{i=1}^N (y_i - A - Bx_i)^2} \quad (1.16)$$

1.4.2 Linear Regression Using Excel

The linear regression formulas for calculating A , B , σ_y , σ_A , σ_B and also R^2 can be obtained using the Excel worksheet function **LINEST** (Line Estimation). Enter the x data, y data, and the error on y (if available) into columns in an Excel worksheet.

The **LINEST** function in Excel is used to perform linear regression and obtain related statistics. With this function, you will get several parameters such as the intercept, slope, and other statistics related to the goodness of fit. It is an array function that returns multiple outputs at once. The syntax of the **LINEST** function is:

=LINEST(known_y's, [known_x's], [const], [stats])

Parameters:

1. known y's:
This is the range of cells containing the dependent variable, i.e., the data you want to predict or explain. This is a required parameter.
2. known x's:
This is the range of cells containing the independent variable, i.e., the data used for prediction.
3. const (optional):
A logical value that determines how the function should be calculated. If TRUE or omitted: the function calculates the intercept A normally, if FALSE: the function forces the intercept A to be zero.
4. stats (optional):
A logical value that determines the type of data to return. If TRUE: the function returns additional regression statistics, if FALSE or omitted: the function only returns the slope B and intercept A .

	A	B	C	D
1	Pengujian Hukum Hooke			
2				
3	Nomor	m (gram)	y (cm)	Δy (cm)
4		Δm diabaikan		
5				
6	1	200	1.1	0.3
7	2	300	1.5	0.3
8	3	400	1.9	0.3
9	4	500	2.8	0.3
10	5	600	3.4	0.3
11	6	700	3.5	0.3
12	7	800	4.6	0.3
13	8	900	5.4	0.3

Figure 1.4: Example of data entry in an Excel worksheet

As an example, it is shown in Figure 1.4 (from the book *Taylor; An Introduction to Error Analysis, the Study of Uncertainties in Physical Measurements*, page 25). The data for the variable x are entered in cells B6 through B13, the data for the variable y are entered in cells C6 through C13, and the errors in y are entered in cells D6 through D13 (even though these error values are not used in the LINEST formula).

We will determine the values of A , B , σ_y , σ_A , σ_B and also R^2 , which makes a total of six values. Therefore, highlight (block) six cells in the Excel worksheet, for example from cell B19 to C21. Then, in the formula bar, type:

=LINEST(C6:C13,B6:B13, TRUE, TRUE)

then press simultaneously: **Ctrl+Shift+ENTER**. The result is shown in Figure 1.5, where the values inside the box are the

output from LINEST, while the text outside the box is additional, for explanatory purposes. It can be verified through manual calculations that the values produced by the LINEST function can also be obtained using the previously given linear regression formulas.

18		B	A	
19	slope =	6.07E-03	-3.14E-01	= titik potong
20	Δ slope =	3.74E-04	2.23E-01	= Δ titik potong
21	R^2 =	0.98	0.24	= σ_y

Figure 1.5: Results of the LINEST Function